

ILES DE LA MADELEINE, QC, Canada

WMO: 717100

Lat: 47.425N

Lon: 61.775W

Elev: 25

StdP: 14.68

Time Zone: -4.00 (NAA)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	2.2	5.9	-4.9	4.2	3.0	-0.7	5.3	6.7	43.8	18.5	41.1	18.2	14.8	240	0.793

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	9.3	75.4	67.5	73.1	66.4	71.2	65.2	69.3	73.6	67.8	71.6	66.3	70.0	13.8	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.7	101.7	71.9	66.2	96.7	70.4	64.7	91.8	69.0	33.5	73.6	32.2	71.7	31.1	70.0	76.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 36.3	31.6	28.1	DB	-1.2	80.6	5.3	2.3	-5.0	82.3	-8.1	83.7	-11.1	85.0	-15.0	86.7	(4)
(5)			WB	-2.3	72.0	5.3	1.6	-6.1	73.2	-9.2	74.2	-12.2	75.1	-16.1	76.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	42.4	23.0	20.6	26.2	34.8	44.6	54.3	64.0	65.0	58.1	48.0	38.8	30.0	(6)	
	DBStd	16.27	7.81	8.35	7.20	5.20	5.61	5.87	4.54	4.19	5.09	5.45	6.23	6.32	(7)	
	HDD50	4132	836	823	739	457	189	22	0	0	3	103	339	621	(8)	
	HDD65	8348	1301	1243	1204	906	632	325	73	52	214	526	786	1086	(9)	
	CDD50	1362	0	0	0	0	21	150	435	466	245	42	3	0	(10)	
	CDD65	103	0	0	0	0	0	2	43	52	6	0	0	0	(11)	
	CDH74	133	0	0	0	0	0	5	55	68	5	0	0	0	(12)	
	CDH80	6	0	0	0	0	0	0	2	4	0	0	0	0	(13)	
	WSAvg	14.3	17.2	15.8	15.2	14.2	12.8	12.1	11.5	11.6	13.4	14.9	16.0	17.4	(14)	
	PrecAvg	41.3	4.2	3.0	3.5	3.1	2.6	2.6	2.9	3.0	3.4	4.0	4.1	4.5	(15)	
(16) Precipitation	PrecMax	54.7	8.7	6.3	9.1	5.7	4.7	5.0	7.8	5.7	5.7	7.0	7.1	8.1	(16)	
	PrecMin	31.9	1.7	0.9	1.6	0.9	1.0	0.5	0.8	1.0	1.3	1.5	1.4	2.2	(17)	
	PrecStd	5.9	1.7	1.5	1.6	1.5	1.0	1.1	1.7	1.3	1.3	1.5	1.3	1.5	(18)	
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	44.9	42.1	43.6	53.6	65.9	73.7	78.6	79.7	73.9	64.6	57.2	48.9	(19)	
		MCWB	44.5	41.4	41.9	49.2	56.6	64.3	69.7	70.1	67.6	60.5	55.5	48.0	(20)	
	2%	DB	40.2	37.5	39.9	48.5	60.4	69.2	75.4	75.6	70.1	60.8	53.8	44.5	(21)	
		MCWB	39.5	36.5	38.2	45.3	53.0	61.6	67.5	67.9	64.8	57.3	52.3	43.5	(22)	
	5%	DB	36.3	34.2	37.0	44.9	56.4	66.1	73.2	73.4	67.6	58.4	50.9	41.3	(23)	
		MCWB	35.2	33.2	35.3	42.0	50.8	59.7	66.4	66.6	63.1	55.7	49.3	40.2	(24)	
	10%	DB	33.4	31.7	34.8	41.7	53.0	63.1	71.1	71.6	65.6	56.1	47.7	38.7	(25)	
		MCWB	32.4	30.8	33.3	38.8	48.3	57.7	65.1	65.7	61.6	53.3	46.0	37.5	(26)	
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	44.4	41.7	42.5	50.6	58.5	66.9	71.2	71.3	69.0	61.9	56.0	48.0	(27)	
		MCDB	44.7	42.0	43.2	52.3	64.0	71.6	76.5	76.2	72.6	63.8	56.8	48.6	(28)	
	2%	WB	39.7	36.7	38.6	46.1	54.7	63.4	69.4	69.5	66.2	58.2	52.6	43.6	(29)	
		MCDB	40.1	37.4	39.5	47.8	58.8	67.0	73.6	73.9	69.0	59.9	53.6	44.3	(30)	
	5%	WB	35.4	33.4	35.7	42.5	51.6	60.9	67.9	68.1	64.2	55.9	49.6	40.6	(31)	
		MCDB	36.2	34.0	36.7	44.6	55.3	64.9	71.6	72.0	66.8	58.1	50.8	41.3	(32)	
	10%	WB	32.5	30.9	33.4	39.3	48.8	58.4	66.3	66.6	62.3	53.6	46.3	37.7	(33)	
		MCDB	33.3	31.6	34.6	41.2	52.4	62.4	70.1	70.3	65.0	55.9	47.5	38.7	(34)	
(36) Mean Daily Temperature Range	5% DB	MDBR	9.7	10.7	9.3	8.5	10.5	10.8	10.1	9.3	9.0	8.1	7.8	7.6	(35)	
		MCDBR	15.1	14.1	11.9	13.6	16.4	15.1	12.6	11.6	10.9	10.5	12.1	12.1	(36)	
		MCWBR	15.9	14.5	11.5	11.2	11.1	9.8	7.3	7.2	8.2	9.6	12.9	12.9	(37)	
	5% WB	MCDBR	15.1	14.1	12.0	13.1	15.3	13.6	11.2	10.1	10.1	10.3	12.2	12.3	(38)	
		MCWBR	15.9	14.7	12.0	11.9	11.6	10.0	7.6	7.3	8.9	10.2	13.1	13.3	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.305	0.296	0.327	0.351	0.368	0.385	0.409	0.396	0.363	0.353	0.327	0.311	(40)	
	taud		2.389	2.305	2.286	2.421	2.446	2.389	2.389	2.427	2.485	2.453	2.438	2.388	(41)	
	Ebn at Noon		246	275	283	286	284	279	270	269	269	252	235	227	(42)	
	Edh at Noon		22	29	35	34	35	37	36	34	29	26	21	20	(43)	
(44) All-Sky Solar Radiation	RadAvg		283	567	1016	1382	1673	1788	1774	1574	1198	731	352	215	(44)	
	RadStd		25	62	86	116	137	165	143	107	97	64	40	19	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	+0.88	N/A	N/A	N/A	+25	(46)
(47)	Regional (20 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page